

Vision-Language Models as Zero-Shot Pairwise Progress Rewards for Visual Robotic Reinforcement Learning

Leonardo Solidoro

leonardo.solidoro@tum.de

Tien Dung Pham

tien.dung.pham@tum.de

1. Introduction & Motivation

Reinforcement learning often requires either manually specified reward functions or learned reward models from human feedback, both of which can be costly and difficult to obtain [3]. In robotic manipulation, reward design is especially problematic because task progress may depend on task-specific quantities such as object position, gripper-object distance, or goal completion, which are not always available from raw visual observations alone. Therefore, vision-language models can offer a promising alternative by providing a reward signal that generalizes across diverse tasks and visual environments.

Recent work has shown that pretrained vision-language models can be used as zero-shot reward models by scoring visual observations against natural-language task descriptions [3]. Rocamonde et al. [3] implement this idea using CLIP, where the reward is computed from the cosine similarity between an image embedding and a text-goal embedding. However, single-frame image-text similarity may not be sufficient for robotic manipulation tasks where progress is relative (e.g., when comparing whether a drawer is more open than before, subtle differences can trick the model’s judgment). Furthermore, for multi-stage tasks, different stages can look similar and be scored erroneously. (e.g., when making a sandwich, placing the first bread slice and the last can result in similar images when viewed from above).

This project investigates a pairwise alternative. Instead of asking whether one image looks like the goal, we present a VLM with before/after frame pairs and goal text to produce progress judgments that can be used for ranking rollouts and training an RL agent. We will mainly focus on large-scale vision-language models, such as Qwen3-VL [4], in this zero-shot pairwise configuration to evaluate whether their increased model capacity and large-scale pretraining translate to better progress reward judgments.

2. Research Question

The main research question is: Can a frozen vision-language model judge manipulation task progress from be-

fore/after image pairs well enough to provide a useful reward signal for reinforcement learning?

More specifically, this project asks:

1. Can pairwise VLM rewards correlate with Meta-World ground-truth rewards and success metrics?
2. Can pairwise VLM rewards rank successful rollouts above failed or partially successful rollouts?
3. Can an RL agent trained with pairwise VLM rewards achieve task success rate similar to ground-truth reward signals?
4. How do large-scale VLMs (e.g., Qwen3-VL) compare to smaller-scale or vision-only alignment models (e.g., CLIP) in providing rewards for manipulation tasks?
5. Where do VLM judgments break down, such as visually similar states, subtle differences in spatial positioning, long-horizon multi-stage tasks, or complex tasks?

3. Proposed Methodology

We propose a pairwise VLM progress reward. Given two rendered frames from a rollout, I_t and I_{t-k} , and a language goal g , the VLM predicts whether the second frame shows progress toward the goal compared to the first frame. The proposed reward is:

$$r_t^{VLM} = VLMProgress(I_{t-k}, I_t, g) \quad (1)$$

The VLM output will be converted into a numerical reward using the scale:

$$r_t^{VLM} \in \{0, 1, 2, \dots, 10\} \quad (2)$$

The reward format is a discrete progress score, motivated by RoboReward’s use of 0-5 progress labels and its finding that progress-based rewards are more useful than binary success rewards in their RL experiments [2].

Our pairwise implementation will focus on testing state-of-the-art vision-language models such as Qwen3-VL to evaluate the impact of large-scale pretraining on progress judgment accuracy.

4. Experimental Setup

The project will use Meta-World [5] as the simulation benchmark. Meta-World contains 50 simulated manipulation tasks and uses a shared Sawyer robot setup across tasks.

The initial task set will be:

- Basic Manipulation: reaching, pushing, pick-and-place.
- Object Interaction: Opening/closing doors and drawers, pressing buttons.
- Complex Tool Use: Inserting pegs into holes, using a hammer, picking objects out of bins.

For each task, rollouts will be collected from expert policies, near-expert policies (rollouts with added noise or early stopping to simulate near-misses and progress), random policy (rollouts from a random explorer to provide examples of failure and lack of progress)

4.1. RL Training Plan

The second stage of the project will train an RL agent using Soft Actor-Critic (SAC) [1] and substituting the Meta-World rewards with VLM rewards.

The strategy for computing VLM rewards efficiently during training, mitigating VLM inference costs, will involve two approaches:

1. Periodic Reward Computation: Rewards are computed only every k steps. (e.g., $k = 5$). The VLM will compare the current frame I_t with a frame from the recent past I_{t-k} . For steps within the window, the reward will be held constant or linearly interpolated.
2. Frame Selection: For multi-stage tasks, we propose a heuristic where the reference frame I_{t-k} is only updated upon detecting significant sub-goal completion. Implementation could involve using the VLM itself to periodically check if a sub-goal description (e.g., "grasping the handle") has been satisfied before advancing the reference frame for the next phase (e.g., "pulling the drawer").

5. Evaluation

We will compare the following reward signals:

1. Meta-World ground-truth reward: Exact mathematical reward functions for each task [5]. This reward will serve as the reference or upper baseline.
2. Single-frame CLIP reward: Image-text similarity baseline [3].
3. Pairwise VLM progress reward: The proposed method. We will specifically evaluate the performance of Qwen3-VL-8B in this pairwise configuration to determine if its increased model capacity and large-scale pretraining translates to better rewards.

5.1. Offline Reward Evaluation

First, the reward signals will be evaluated before full RL training. The offline metrics will be:

- MAE between predicted rewards and Meta-World ground-truth rewards
- Rollout ranking accuracy

5.2. RL Policy Evaluation

After training, policies will be evaluated using the true Meta-World success metrics:

- Success Rate: Fraction of episodes where the agent achieves the task goal within the environment's time limit. Meta-World defines specific binary success criteria for each task, typically involving whether a target object has reached a goal region within a predefined tolerance (e.g., within 5cm of the target position).
- Sample Efficiency: Average return vs. number of environment steps.

6. Expected Contribution

This project will evaluate whether pairwise VLM progress judgments can serve as reward signals. The expected outputs are:

1. A pipeline for scoring Meta-World frame pairs with a VLM-based progress reward.
2. A comparison between Meta-World rewards, CLIP-style rewards, and pairwise VLM rewards.
3. RL training results with different rewards, using SAC as the training algorithm.
4. Failure analysis on cases such as visually similar states, subtle differences in spatial positioning, long-horizon multi-stage tasks and complex tasks.

References

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